

When High Accuracy Misleads in Literature-Derived Machine Learning for Deep Eutectic Solvent Recommendation

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ABSTRACT

Deep eutectic solvents (DESs) are widely used in analytical sample preparation, yet selecting suitable systems for specific analytes remains challenging due to the large combinatorial design space and reliance on empirical screening. Machine learning (ML) has been proposed as a data-driven alternative, but its reliability under literature-derived data constraints is unclear. In this study, an experiment-level dataset for DES-based pesticide extraction (757 records from 132 studies, organized into 626 (DOI, analyte) groups) was reconstructed and evaluated using a leakage-controlled, DOI-grouped validation framework. Following a comprehensive leakage audit, the modeling pipeline was rebuilt using a split-first, training-only strategy with leakage-clean descriptors. The hybrid model, combining global classification and pairwise preference learning, improved record-level classification performance (ROC-AUC, MCC) relative to the baseline, but did not improve group-level ranking. In restricted comparable multi-candidate groups, ranking performance decreased and the hybrid model frequently underperformed, with no statistically significant advantage observed. Sanity baseline analysis further showed that performance was only modestly separated from simple heuristic selection, indicating that much of the predictive signal reflects coarse descriptor-level compatibility. These results are explained by sparse comparative structure, study-bounded target definition, and limited descriptor representation. Overall, this work demonstrates that in literature-derived DES extraction settings, improvements in predictive accuracy do not necessarily translate into reliable recommendation capability, highlighting the need for data-centric evaluation and improved experimental reporting for ML-driven chemical recommendation.

KEYWORDS. Deep eutectic solvents; Machine learning; Information leakage; Ranking systems; Data-centric modeling; Analytical chemistry

1 INTRODUCTION

Deep eutectic solvents (DESS) have emerged as versatile and environmentally favorable alternatives to conventional organic solvents in analytical chemistry, particularly in sample preparation and extraction workflows. Their tunable composition—defined by combinations of hydrogen bond acceptors (HBAs), hydrogen bond donors (HBDs), and molar ratios—enables control over key physicochemical properties such as polarity, viscosity, and hydrogen-bonding capacity. As a result, DES-based systems have been widely explored for the extraction of organic contaminants, including pesticides, across diverse matrices.^{1,2}

Despite these advantages, selecting an appropriate DES system for a given analyte remains a non-trivial task. The design space is inherently combinatorial, yet experimental studies typically evaluate only a limited number of candidate systems, often focusing on structurally related DESS or narrow compositional variations.^{3,4} This constrained exploration limits coverage of the underlying chemical space and results in datasets that are sparse, highly structured, and often biased toward locally optimized solutions.

In response, machine learning (ML) approaches have been increasingly proposed to accelerate DES design and selection. Most existing studies focus on predicting physicochemical properties such as density, viscosity, or solubility, where relatively structured datasets are available.^{5,6} In contrast, the use of ML for recommending DES systems for specific analytical tasks—particularly extraction of target analytes—remains limited and conceptually more challenging.⁷ However, whether such approaches can support reliable recommendation under realistic, literature-derived data constraints remains an open question.

A key assumption underlying literature-driven ML approaches is that experimental data reported in published studies provide sufficient supervision for learning transferable relationships. However, literature-derived datasets differ fundamentally from designed

datasets: they are generated through independent, study-specific optimization processes, typically aimed at identifying a single best-performing system rather than systematically characterizing performance across competing alternatives. As a consequence, such datasets may contain abundant information about clearly suboptimal conditions but limited information about relative differences among near-optimal candidates.

From a ML perspective, this leads to uneven and context-dependent supervision. While coarse discrimination between optimal and non-optimal systems may be learnable, the fine-grained comparisons required for recommendation—i.e., ranking multiple candidates within the same decision context—may be weakly supported. In addition, experimental outcomes are inherently conditioned by study-specific factors such as matrix, extraction conditions, and candidate selection, which are not always consistently reported or fully represented in the available descriptors.^{8,9}

An additional methodological concern is the potential for information leakage in literature-derived modeling pipelines.¹⁰ Variables derived from experimental outcomes—such as recovery, enrichment factor, or performance-based indicators—may inadvertently be incorporated into the feature space, leading to overly optimistic estimates of model performance. Pairwise or delta-based feature constructions are particularly susceptible to this issue if outcome-related information is not strictly excluded.¹¹

Finally, evaluation of recommendation models requires careful alignment between metrics and the structure of the underlying data. Standard record-level metrics such as ROC–AUC or MCC assess global classification performance, but do not directly capture the ability of a model to make context-specific decisions among competing candidates. In recommendation settings, the relevant unit of evaluation is the set of alternatives defined within a given experimental context, rather than individual records considered in isolation.

In this work, we critically examine the feasibility of literature-driven ML for DES recommendation in analytical extraction. Rather than focusing on maximizing predictive performance, our objective is to assess whether the structure and informational content of literature-derived datasets support meaningful recommendation behavior. Specifically, we (i) reconstruct an experiment-level dataset from DES-based pesticide extraction studies, (ii) develop a leakage-aware modeling framework, and (iii) evaluate model behavior under group-consistent validation settings that reflect realistic experimental contexts.

Within the present literature-derived DES extraction setting, this study adopts a data-centric perspective, arguing that model performance depends not only on algorithmic design but also on how experimental data are generated, structured, and reported.¹²

2 METHODOLOGY

The overall workflow of the data-driven framework, from literature extraction to leakage-clean model evaluation, is summarized in Figure 1.

2.1 Literature Search and Data Extraction

A structured literature search was conducted to identify studies reporting the use of DESs for pesticide extraction in analytical workflows. After duplicate removal and screening, 132 studies were retained for experiment-level data extraction.

Experiment-level data were extracted using a structured AI-assisted workflow designed to capture all reported DES systems within each study, including optimal and suboptimal conditions, without aggregation or inference. Each DES–analyte–condition combination was treated as an independent record.

To assess extraction reliability, a subset of studies was randomly selected for manual validation, and key experimental fields were cross-checked against the original publications.

Additional details required for reproducible screening, including the search query, eligibility criteria, extraction protocol, validation audit, and prompting strategy are provided in the Supporting Information (Text S1–S3 and Table S1).

2.2 Dataset Construction and Target Definition

The initial extraction yielded 956 experiment-level records, of which 757 were retained after data cleaning and consistency filtering. All chemical components were standardized and mapped to canonical SMILES representations, and each DES system was defined as a unique combination of HBA, HBD, and molar ratio.

Experimental records were organized into competitive groups defined at the (DOI, analyte) level, such that each group contains all DES systems evaluated for a given analyte within a single study. This grouping ensures that comparisons are performed within a consistent experimental context.

The binary target variable, *is_optimal*, was defined within each group. DES systems achieving the maximum reported recovery within the same (DOI, analyte) group were labeled as optimal (*is_optimal* = 1), while all other systems were labeled as non-optimal (*is_optimal* = 0). In cases of ties, multiple systems were assigned the optimal label.

Accordingly, the target should not be interpreted as an intrinsic chemical optimum, but as the best observed condition within a study-specific candidate set. This target is therefore conditional on the experimental design, candidate coverage, and reported context of each source study. The learning problem is thus relative and study-bounded by construction. This formulation should be interpreted as an operational target definition for the present study rather than as a unique or universal representation of chemical optimality. Alternative target formulations, such as top-k labels or direct recovery prediction, may yield different model behavior and are outside the scope of the current analysis. Because the target variable is defined

conditionally within each (DOI, analyte) group, global feature–target correlation analysis was not considered sufficiently informative for the aims of this study and could be misleading outside the group context.¹³

2.3 Descriptor Calculation and Feature Engineering

Molecular descriptors were computed from canonical SMILES representations using RDKit. A set of interpretable physicochemical descriptors was selected, including molecular weight, octanol–water partition coefficient (logP), topological polar surface area (TPSA), hydrogen-bond donor and acceptor counts, number of rotatable bonds, ring count, and heavy atom count.¹⁴

Descriptors were calculated for HBA, HBD, and analyte components, and subsequently combined to generate both DES-level aggregate descriptors and DES–analyte interaction features. DES-level descriptors were constructed using composition-weighted representations based on molar ratios, while interaction features were derived using difference, ratio, and product transformations to capture relative compatibility between DES systems and analytes.

The mathematical definitions of all engineered features and their physicochemical rationale are provided in the Supporting Information (Text S4 and Tables S2 and S3).

2.4 Limitation of the descriptor space. This feature space was intentionally restricted to leakage-clean, interpretable, descriptor-based variables. It does not explicitly encode several experimentally important determinants of extraction performance, such as matrix effects, pH, water content, ionic strength, or process-specific conditions, unless such information is indirectly reflected in the reported records.

2.5 Feature Refinement and Leakage Control

To ensure valid model evaluation, strict leakage control was applied. All outcome-related variables, including recovery, enrichment factor, limit of detection (LOD), and any derived

indicators, were excluded from the feature space prior to model training. The excluded variables and their rationale are summarized in Table 1.

Additional preprocessing steps included removal of features with high missingness, near-zero variance, and excessive collinearity. A detailed summary of retained and excluded features at each preprocessing stage is provided in Table S4 (Supporting Information).

All preprocessing and feature selection steps were performed exclusively on the training data within each split to prevent information leakage from the evaluation set.¹¹

A split-wise pseudocode description of preprocessing, feature filtering, pairwise construction, and inference is provided in Text S8 to ensure full transparency of the pipeline implementation. Feature scaling was not expected to materially affect the conclusions. While certain models such as logistic regression may benefit from normalization, tree-based models are inherently scale-invariant. Given that the primary focus of this study is on dataset structure and evaluation validity rather than model optimization, scaling was not considered a critical factor.^{15,16}

2.6 Modeling Strategy

Two complementary modeling approaches were considered: a global classification model and a pairwise preference model. The global model estimates the probability that a DES system is optimal within its corresponding (DOI, analyte) group, while the pairwise formulation captures relative preferences between competing systems.

Baseline models, including Logistic Regression, Random Forest, and XGBoost, were evaluated to establish a reference for model performance. Model configurations and hyperparameters are provided in Table S5.

Pairwise preference learning was implemented by constructing feature differences between DES systems within the same group. A hybrid model was defined by combining the global classification probability with aggregated pairwise preference scores, as described in the

Supporting Information (Text S5). In this study, the pairwise component is not assumed to improve performance a priori, but is evaluated as a test of whether limited comparative supervision can provide additional ranking signal beyond the global model.

The pairwise component is expected to provide complementary but constrained information relative to the global model. In the revised interpretation, the pairwise branch should be viewed primarily as a falsification-oriented probe of whether limited comparative supervision adds ranking value beyond the global model.

2.7 Validation and Evaluation Protocol

A DOI-grouped validation strategy was adopted to approximate realistic generalization across studies, such that all records from the same publication were assigned to the same fold or held-out set. This design prevents study-level information leakage and provides a more reliable estimate of model transferability.¹¹

Four complementary evaluation blocks were used in this study: (i) baseline DOI-grouped cross-validation across model classes, (ii) a validation-strategy comparison designed to illustrate performance inflation under random row-wise splitting, (iii) a final leakage-clean DOI-held-out comparison between the baseline full model and the hybrid formulation, and (iv) sanity baseline evaluation using repeated random within-group ranking and a simple leakage-clean heuristic. These analyses serve different purposes and should not be interpreted interchangeably. A schematic summary of these evaluation blocks, including dataset versions, feature spaces, split strategies, and reporting locations, is provided in Table 2.

Model performance was evaluated at both the record and group levels. Record-level classification was assessed using receiver operating characteristic area under the curve (ROC-AUC), precision-recall area under the curve (PR-AUC), Matthews correlation coefficient (MCC), balanced accuracy, precision, recall, and F1-score.¹⁴ Group-level recommendation

performance was evaluated within each (DOI, analyte) group using Hit@1, Hit@3, and mean reciprocal rank (MRR).

For the hybrid model, ranking was performed using pairwise inference within each group with aggregation of pairwise comparisons, as detailed in the Supporting Information. Given the known sparsity of competitive group structures in literature-derived datasets, additional restricted evaluations on comparable multi-candidate groups were performed to assess ranking performance under more meaningful decision contexts.

To contextualize learned model performance against trivial and simple rule-based alternatives, two sanity baselines were additionally evaluated at the group level: repeated random within-group ranking and a deterministic leakage-clean heuristic based on analyte–DES polarity mismatch.

2.8 Implementation and Reproducibility

All analyses were implemented in Python using RDKit, scikit-learn, XGBoost, pandas, and numpy. The software environment and library versions are reported in Table S14.

The Supporting Information provides detailed methodological and computational documentation, including literature search and extraction protocols (Text S1–S3), feature definitions (Text S4), hybrid model formulation (Text S5), and pairwise dataset characterization (Text S6–S7), together with the curated dataset (Dataset S1).

To make train/test isolation explicit, the revised Supporting Information now includes split-wise pseudocode for preprocessing, pairwise construction, and within-group ranking inference.

The provided code (Code S1) reproduces the leakage-clean data processing, model training, and evaluation workflows reported in this study. While it ensures reproducibility of the final results, it is not intended as a formal proof of the absence of all possible indirect dependencies beyond the explicitly controlled pipeline steps.

3 RESULTS

3.1 Dataset Structure and Target Definition

The reconstructed dataset comprised 757 experiment-level records from 132 studies, organized into 626 (DOI, analyte) competitive groups. Each group defines a local comparison space in which DES systems were evaluated for the same analyte within a single study. Despite its apparent size, the dataset exhibited highly uneven informative content. Using a Δ Recovery threshold of 5% as an operational diagnostic of meaningful competition, only 21 groups (~3.4%) were classified as informative, while the remainder were dominated by a single optimum. Qualitative sensitivity to alternative thresholds is shown in Figure S1.

This structure indicates that the dataset is dominated by coarse discriminative signal—clear separation between optimal and suboptimal systems—while fine-grained ranking signal is scarce. Consequently, the effective sample size for learning relative performance differences is substantially smaller than the nominal number of records. The target variable, *is_optimal*, was defined within each (DOI, analyte) group based on maximum reported recovery. This definition is inherently contextual and study-specific, reflecting the best-performing system among those tested rather than an absolute chemical optimum. In rare cases, multiple systems shared the optimal label due to ties. Thus, label validity depends on both observed performance and the completeness of the candidate set explored within each study.

These properties imply that the learning problem is intrinsically local and that limited exploration of near-optimal regions restricts informative comparisons between competing systems. This is further supported by the distribution of recovery gaps within groups (Figure 4), where most groups show large differences between the best and second-best systems. Taken together, the dataset contains abundant coarse signal but limited information for reliable ranking or recommendation, constraining both statistical power and interpretability of positive labels as transferable optima. The observed class imbalance is an intrinsic consequence of the

group-based target definition, where typically a single system is labeled as optimal within each group, and is therefore treated as a structural property rather than a modeling artifact.^{17,18}

3.2 Baseline Modeling under DOI-Grouped Validation

To establish a realistic reference point, baseline classifiers were evaluated under DOI-grouped cross-validation, in which all records from the same study were assigned to the same fold. This design minimizes study-level information leakage and provides a more stringent estimate of generalization than random row-wise splitting.¹¹

Under DOI-grouped validation, all baseline models showed modest and consistent performance (Table 3). ROC-AUC ranged from 0.5657 to 0.5845, while MCC remained near zero (−0.0404 to 0.0238).¹⁹ Balanced accuracy was also near chance level (0.4845–0.4964), indicating limited discrimination under leakage-controlled conditions.

Although PR-AUC values were relatively high, this reflects the prevalence of the positive class under the within-group target construction and should not be interpreted as evidence of decision-level usefulness. MCC and balanced accuracy provide a more conservative assessment of model performance, and fold-level variability (Table S10) confirms that this pattern is consistent across models.

Comparison with random row-wise splitting highlights the impact of validation design. Performance is strongly inflated under random splitting (Figure 2; Table S11), confirming the presence of study-level leakage and reinforcing the need for grouped validation.¹¹

Overall, these results indicate that baseline performance is constrained primarily by dataset structure rather than model choice, motivating the leakage-clean reconstruction described below.

3.3 Leakage Audit and Pipeline Reconstruction

Inspection of the original pipeline revealed that the pairwise feature space included outcome-derived variables such as recovery, enrichment factor, and LOD, which are not available at prediction time and introduce information leakage (Table 1).

To address this issue, the pipeline was reconstructed using a strictly leakage-clean feature definition. All outcome-related variables were excluded, and only chemically interpretable descriptors were retained (Table S15). The revised workflow (Figure 5) follows a split-first, training-only paradigm.

Under this corrected framework, previously observed performance gains were not reproduced. Instead, results indicate a more constrained outcome: modest improvement in record-level classification but no improvement in recommendation performance. These conclusions remain bounded by the limited pairwise sample size and the restricted descriptor regime.

3.4 Leakage-Clean Holdout Evaluation with True Pairwise Inference

Under DOI-held-out evaluation, the hybrid model showed moderate improvement in record-level classification relative to the baseline (Table S16), indicating improved global discrimination under leakage-controlled conditions. However, this did not translate into improved recommendation performance. At the group level, ranking metrics (Hit@1, MRR) were comparable or slightly lower for the hybrid model across the full holdout set (Table 4).

When evaluation was restricted to comparable multi-candidate groups, performance decreased substantially (Table S7), and the hybrid model did not outperform the baseline, showing degraded ranking in several cases. This pattern was consistent across both ≥ 2 and ≥ 3 candidate subsets.

Group-wise statistical comparison confirmed the absence of any consistent advantage of the hybrid model (Text S9; Table S9). Given the limited number of evaluable groups, differences remained within sampling variability.

These results reflect the structure of the leakage-clean pairwise dataset (Table S6), which is small, highly structured, and derived from a limited number of informative groups. Under these conditions, pairwise learning captures coarse directional preferences but does not provide stable ranking improvements.

3.5 Sanity Baselines for Group-Level Ranking

To contextualize the practical meaning of the observed ranking performance, two sanity baselines were evaluated under the same leakage-clean DOI-held-out setting: repeated random ranking within each (DOI, analyte) group and a simple leakage-clean heuristic based on analyte–DES polarity mismatch. The random baseline provides a lower-bound reference corresponding to the absence of learned structure. The ML baseline substantially outperformed random ranking across all evaluation scopes (Table 6), confirming that the observed performance is not purely trivial.

The heuristic baseline provides a more stringent and chemically interpretable reference. This heuristic captures simple compatibility patterns already encoded in the descriptor space. In comparison, the ML model showed only limited separation from the heuristic baseline in comparable-group subsets (Table 6 and Table S17), indicating that a substantial fraction of the observed ranking performance can be explained by simple rule-based structure.

These sanity checks demonstrate that while ML models learn non-trivial patterns beyond random behavior, their advantage over simple physicochemical heuristics remains limited under the present data structure.

3.6 Group-Structure Diagnostics and Evaluation Fragility

Analysis of the evaluation space reveals that most (DOI, analyte) groups do not constitute meaningful ranking problems. As shown in Table 5, the median group size is one, and the majority of groups contain a single optimal system. This structural sparsity directly affects

ranking metrics. Measures such as Hit@1 and MRR assume multiple competing candidates, and when this assumption is violated, metrics become inflated and lose interpretability.^{9,11}

This effect is evident when restricting evaluation to comparable multi-candidate groups, where performance decreases substantially (Table S7). Under these conditions, the hybrid model does not outperform the baseline and often performs worse.

Uncertainty analysis further shows that estimates are statistically unstable (Table S8), with wide and overlapping confidence intervals. This indicates that observed differences are not robust. In this study, evaluation fragility is evidenced by the divergence between full-group and restricted evaluation, combined with instability due to the limited number of evaluable groups.

3.7 Implications for Recommendation Validity

The results demonstrate that improvements in predictive performance do not necessarily translate into meaningful recommendation capability.^{13,20,21} While the hybrid model improves record-level classification (Table S16), it does not produce consistent gains in ranking performance (Table 4; Table S7). Sanity baseline comparisons are consistent with this conclusion, indicating that performance is only modestly separated from simple heuristic alternatives in comparable-group subsets.

These findings indicate that much of the observed performance arises from coarse compatibility patterns rather than robust ranking of near-optimal candidates. Under the present data structure, distinguishing among high-performing alternatives remains unresolved. More broadly, these results highlight that data structure, target conditionality, and limited comparative supervision are the primary constraints on recommendation performance in literature-derived datasets.

4 DISCUSSION

4.1 From Model Performance to Data-Structure Constraints

The results indicate that model performance is constrained less by algorithmic choice than by the structure and informational content of the underlying data. Under DOI-grouped cross-validation, all baseline models exhibit modest and consistent performance, with ROC–AUC values near 0.56–0.58 and near-zero MCC, reflecting limited discriminative power once study-level leakage is controlled.¹⁹

Following reconstruction under strictly leakage-clean conditions, the hybrid model shows moderate improvements in record-level classification metrics (Table S16). However, these gains do not translate into improved recommendation capability. Across evaluation settings, including restricted comparable-group subsets, the hybrid model does not outperform the baseline and, in several cases, exhibits degraded ranking performance (Table S7).

Sanity baseline comparisons (Section 3.5; Table 6; Table S17) further contextualize these results. While the ML baseline clearly outperforms random ranking, its advantage over a simple physicochemical heuristic is limited in comparable-group subsets, indicating that much of the predictive signal arises from coarse compatibility patterns already encoded in the descriptor space.⁹ This discrepancy reflects a fundamental limitation of the available supervision. The dataset contains sufficient signal to distinguish clearly suboptimal systems from locally optimal ones, but lacks the competitive structure required to learn stable ranking relationships among high-performing alternatives. As shown by group-structure diagnostics (Table 5; Figures 3–4), most groups contain a single candidate or a trivially dominant system.

Under such conditions, increased model complexity or pairwise learning is unlikely to compensate for the absence of informative comparisons. The inability of the hybrid model to improve ranking—even when explicitly designed to capture relative preferences—provides empirical evidence that the primary bottleneck lies in the data structure and available supervision rather than in the modeling framework.

Taken together, these findings support a shift from model-centric optimization toward data-centric analysis. In literature-derived DES extraction settings, dataset structure—not algorithm choice—emerges as the primary determinant of achievable performance.

4.2 Metric–Utility Disconnect in Chemical Recommendation

A central finding is the mismatch between improvements in predictive metrics and practical recommendation utility. Although the hybrid model achieves higher record-level classification performance (Table S16), this does not translate into improved ranking within (DOI, analyte) groups (Table 4; Table S7).

This reflects a fundamental distinction between global classification and local recommendation. Metrics such as ROC–AUC and MCC assess discrimination across the full dataset, whereas recommendation requires reliable ordering within specific decision contexts. A model may improve global separation without improving its ability to resolve competition among high-performing candidates.

The present results follow this pattern. Classification gains primarily reflect improved separation between clearly optimal and clearly suboptimal systems, while fine-grained ranking signal remains limited. When evaluation is restricted to comparable groups, ranking performance decreases and the hybrid model shows no advantage over the baseline.

This observation is consistent with the behavior observed under comparable-group evaluation (Section 3.5, Table 6; Table S17). Although ML models outperform random ranking, their advantage over simple heuristic selection is limited in comparable-group evaluation, indicating that a substantial portion of performance can be explained by simple compatibility rules rather than learned ranking relationships.

This phenomenon can be described as a metric–utility disconnect, where improvements in predictive metrics do not constitute evidence of decision-level usefulness. Similar effects have

been reported in machine learning, where models exploit shortcut structure rather than task-relevant signal.^{13,22}

The implication is that predictive accuracy alone is insufficient for evaluating recommendation models. Reliable assessment requires alignment between evaluation metrics and the structure of the underlying decision problem.

4.3 Limited Role of Pairwise Learning under Sparse Comparative Supervision

Pairwise preference learning is often considered a natural formulation for recommendation problems. In principle, it should be well suited to DES selection, where the objective is to rank competing candidates within a given analyte context. However, under leakage-clean conditions and true pairwise inference, the hybrid model does not provide reliable ranking improvements.

The primary limitation is sparse comparative supervision. As shown in Section 3.1, only a small fraction of groups contains meaningful competition, while most are dominated by a single optimal system. Consequently, pairwise comparisons are coarse, imbalanced, and highly structured.

This is reflected in the leakage-clean pairwise dataset (Table S6), which is small, redundant, and constructed from a limited number of informative groups. The learned preference function therefore captures local patterns rather than transferable ranking relationships.

This structural limitation is directly reflected in the observed ranking behavior under comparable-group evaluation (Table S7). Ablation and sensitivity analyses (Table S12; Table S13) further confirm that the pairwise component does not contribute stable additional signal.^{20,21}

These findings indicate that pairwise learning cannot compensate for insufficient comparative structure. Under sparse supervision, it primarily reinforces coarse directional patterns rather than enabling robust ranking among near-optimal candidates.

Accordingly, the pairwise component should be interpreted not as a performance-enhancing mechanism but as a diagnostic probe. Its lack of improvement provides direct evidence that the dataset does not support the intended recommendation task under current conditions.

4.4 Evaluation Fragility in Literature-Derived Datasets

The group-structure diagnostics in Section 3.5 reveal a broader issue: evaluation fragility under sparse and uneven competitive structure.

Ranking metrics such as Hit@1 and MRR assume meaningful competition within each group. However, most groups in the present dataset violate this assumption (Table 5; Figure 3), leading to inflated metrics driven by trivial cases.

This is evident when comparing full-group and restricted evaluations. While ranking metrics appear high in the full holdout set (Table 4), they decrease under comparable-group restriction, and the hybrid model shows no advantage (Table S7). In several cases, performance is lower.¹¹

Uncertainty analysis (Table S8) shows wide and overlapping confidence intervals, reflecting low statistical power. Rather than supporting model superiority, these results highlight instability in performance estimates.

Sanity baseline comparisons (Section 3.5; Table 6; Table S17) further support this interpretation by showing limited separation from simple heuristic alternatives in comparable-group subsets.

In this study, evaluation fragility is operationally evidenced by:

- (i) divergence between full and restricted evaluations,
- (ii) limited separation from heuristic baselines, and
- (iii) high uncertainty under small sample sizes.

Evaluation fragility thus arises from the interaction between metric assumptions and dataset structure. When these assumptions are violated, performance estimates may overstate model capability.

4.5 Toward AI-Ready Experimental Reporting in Analytical Chemistry

The limitations identified here arise not only from modeling choices, but from how experimental knowledge is generated and reported. Literature-derived datasets often lack competitive structure and systematic reporting of alternatives.

To enable reliable ML-based recommendation, future studies should:

- report near-optimal alternatives alongside optimal systems,
- include low-performing candidates within comparable contexts,
- provide structured experimental metadata (e.g., pH, matrix, conditions),
- design experiments that sample competitive regions rather than single optima.

These practices would directly improve the availability of ranking-relevant supervision.

4.6 Chemical Representation Limits of the Present Leakage-Clean Feature Space

The feature space used in this study is intentionally restricted to leakage-clean, interpretable descriptors. While this ensures methodological validity, it limits representation of key experimental factors such as matrix effects, pH, and process conditions. Consequently, the observed performance ceiling reflects both data structure and representation limits. The model captures compatibility patterns within a constrained descriptor regime, rather than a complete physicochemical system¹³. Under this representation regime, the model is expected to primarily capture coarse descriptor-level compatibility patterns rather than fine-grained physicochemical mechanisms governing extraction performance.^{23,24}

Accordingly, these results should not be interpreted as a general failure of machine learning, but as a limitation of applying leakage-clean descriptor-based models to literature-derived datasets with incomplete representation.

4.7 Target Conditionality and Label Pathology

The target variable reflects the best observed condition within a study-specific candidate set rather than an intrinsic optimum. This introduces label conditionality: the positive label depends on candidate coverage and experimental context.

As a result, the learning task is inherently study-bounded. Predictions cannot be interpreted as transferable chemical optima but as reconstructions of relative preferences under incomplete and context-dependent information.

4.8 Study Boundaries and Limitations

The conclusions of this study are bounded by four conditions: (i) a single application domain (DES-based pesticide extraction), (ii) a study-bounded target definition, (iii) a leakage-clean but chemically restricted descriptor space, and (iv) limited comparable-group sample size. Accordingly, these results should be interpreted not as a general failure of ML or pairwise learning, but as a cautionary assessment of what can be reliably inferred under the present data, target, and evaluation regime. Artificial class balancing techniques such as SMOTE were not applied in this study, as they may disrupt the group-based structure of the dataset and introduce synthetic samples not grounded in experimental comparisons.^{17,25} In this context, class imbalance is not treated as a problem to be corrected, but as a defining characteristic of the underlying data generation process.

5 CONCLUSION

This study evaluated the feasibility of machine-learning-based recommendation of deep eutectic solvent (DES) systems using literature-derived experimental data under a strictly leakage-clean and group-aware evaluation framework.

The results demonstrate that improvements in record-level predictive performance do not necessarily translate into meaningful recommendation capability. Although the hybrid model combining global classification and pairwise preference learning achieved moderate gains in classification metrics under leakage-clean DOI-held-out evaluation, these gains did not yield robust improvement in group-level ranking. Across restricted comparable multi-candidate groups, the hybrid model showed no consistent advantage over the baseline and, in several cases, exhibited degraded performance. Statistical analysis further confirmed that observed differences were not significant under the limited evaluation sample size.

Comparison with sanity baselines provides additional context for interpreting these findings. While ML models outperform random ranking, their advantage over simple leakage-clean physicochemical heuristics is limited in comparable-group evaluation. This indicates that a substantial portion of the observed predictive signal can be explained by coarse compatibility patterns already encoded in the descriptor space, rather than by robust learning of transferable ranking relationships.

These findings indicate that the primary limitation of the problem lies not in model capacity, but in the structure and informational content of the available data. The dataset contains sufficient signal to distinguish clearly suboptimal systems, but lacks the competitive structure required to learn stable ranking relationships among high-performing alternatives. The limited size, redundancy, and structural imbalance of the pairwise dataset further constrain the effectiveness of pairwise learning.

Importantly, the target variable in this study reflects the best observed DES condition within study-specific candidate sets rather than an intrinsic chemical optimum. As a result, the learning task is inherently conditional and bounded by the experimental design and reporting scope of the underlying studies. Under such conditions, improvements in predictive accuracy cannot be interpreted as evidence of transferable or decision-relevant recommendation performance.

Within the present literature-derived DES extraction setting, this work highlights a fundamental challenge in evaluating ML for chemical recommendation. Reliable evaluation requires alignment between data structure, target definition, descriptor representation, and evaluation protocol. When competitive sets are sparse or trivial, standard performance metrics may provide misleading indications of model capability, giving rise to evaluation fragility.

From a practical perspective, the results suggest that progress in AI-assisted DES recommendation will depend less on increasing model complexity and more on improving the structure and reporting of experimental data. In particular, systematic inclusion of near-optimal alternatives, expanded coverage of candidate systems, and consistent reporting of experimental and physicochemical conditions are essential for enabling meaningful recommendation learning.

Overall, this study provides a reproducible, leakage-aware framework for critically assessing recommendation claims in literature-derived datasets and emphasizes the need for data-centric approaches in the development of ML methods for analytical chemistry. These conclusions are therefore explicitly bounded to literature-derived DES extraction datasets under the present target definition, descriptor regime, and evaluation framework, and should not be interpreted as a general limitation of ML across all chemical recommendation settings.

ASSOCIATED CONTENT

Supporting Information

A listing of the contents of each file supplied as Supporting Information is provided below.

The following files are available free of charge.

- Supporting Information: detailed description of literature search and data extraction procedures (Text S1–S3), feature engineering and descriptor definitions (Text S4; Tables S2–S3), hybrid model formulation (Text S5), pairwise dataset structure (Text S6; Table S6), evaluation framework and robustness analyses including ablation and sensitivity studies (Text S7; Tables S7–S13), leakage-clean feature space and classification performance (Tables S14–S16), and sanity baseline evaluation (Text S10; Table S17) (PDF).
- Dataset S1: curated experiment-level dataset of DES-based pesticide extraction records used in this study (CSV).
- Code S1: reproducible Python workflow for leakage-clean preprocessing, model training, evaluation, and sanity baseline analysis (IPYNB).

Author Contributions

Hakim Faraji conceived the study, designed the methodology, developed the computational pipeline, performed data analysis, and wrote the original manuscript. Ruth Rodríguez-Ramos contributed to literature data collection, dataset organization, and preliminary data curation. Juan A. Mendez-Perez provided methodological guidance, contributed to conceptual development, and critically reviewed the modeling framework and manuscript. Bárbara Socas-Rodríguez and Antonio V. Herrera Herrera performed manual validation of extracted data, cross-checking experimental records against original publications, and contributed to data quality assurance. All authors contributed to the interpretation of the results, reviewed the manuscript, and approved the final version.

Funding sources

Research funded through the Talentum program by the Cabildo Insular de Tenerife in collaboration with the Universidad de La Laguna.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

ACKNOWLEDGMENT

The author would like to extend profound thanks to Prof. Miguel Ángel Rodríguez-Delgado for his invaluable guidance, insightful suggestions, and continuous support throughout the duration of this study.

Data and Software Availability

The data underlying this study are available in the published article, the Supporting Information (Dataset S1). The computational workflow and code used to process the data, train the models, and perform the evaluation are available in the Supporting Information (Code S1).

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Tables:

Table 1. Leakage-prone variables excluded from the revised modeling space.

Feature	Category	Reason for exclusion
all_parse_ok	Parsing / technical flag	Potential data-processing artifact; not chemically meaningful for prediction
analyte_parse_ok	Parsing / technical flag	Potential data-processing artifact; not chemically meaningful for prediction
cluster	Post hoc grouping variable	Not a valid pre-outcome chemical predictor
ef	Outcome-derived metric	Directly linked to experimental performance
hba_descriptor_eligible_phase1	Eligibility / technical flag	Technical preprocessing indicator, not a predictive chemical variable
hba_parse_ok	Parsing / technical flag	Potential data-processing artifact
hbd_descriptor_eligible_phase1	Eligibility / technical flag	Technical preprocessing indicator, not a predictive chemical variable
hbd_parse_ok	Parsing / technical flag	Potential data-processing artifact
is_optimal	Target-related label	Direct target leakage
lod	Outcome-derived metric	Directly linked to analytical performance
ratio_parse_ok	Parsing / technical flag	Potential data-processing artifact
recovery	Outcome-derived metric	Used in target definition; direct leakage
target	Target-related label	Direct target leakage
was_inferior	Outcome-derived flag	Derived from comparative performance

Table 2. Summary of experimental settings used across evaluation blocks.

Analysis block	Dataset version	Feature space	Split type	Models	Primary purpose	Reported in
Baseline cross-validation	Clean dataset	Leakage-clean descriptors	DOI-grouped CV	LogReg, RF, XGB	Reference model performance under grouped validation	Table 3, Table S10
Validation strategy comparison	Same dataset	Same feature space	Random vs DOI-grouped	LogReg baseline	Quantify inflation due to improper splitting	Table S11
Final holdout evaluation	Clean dataset	Leakage-clean descriptors	DOI-held-out	LogReg (baseline), Hybrid	Assess classification vs recommendation under strict separation	Table 4, Table S16
Restricted comparable-group evaluation	Holdout subset	Same feature space	DOI-held-out (subset)	Baseline vs Hybrid	Evaluate ranking under meaningful competition	Table S7-S9
Ablation and sensitivity	Holdout subset	Same feature space	DOI-held-out	Full-only, pairwise-only, hybrid	Assess contribution of pairwise component	Table S12, Table S13
Sanity baseline evaluation	Holdout subset	Leakage-clean descriptor subset	DOI-held-out	Random ranking; heuristic baseline	Contextualize group-level ranking against trivial and simple rule-based alternatives	Table 6, Table S17

Table 3. Baseline model performance under DOI-grouped cross-validation.

Model	No. of features	ROC-AUC (mean $\pm$ SD)	PR-AUC (mean $\pm$ SD)	MCC (mean $\pm$ SD)	Balanced Accuracy (mean $\pm$ SD)
Logistic Regression	75	0.5689 $\pm$ 0.1685	0.9263 $\pm$ 0.0580	0.0238 $\pm$ 0.1098	0.4964 $\pm$ 0.0978
XGBoost	75	0.5657 $\pm$ 0.1010	0.9263 $\pm$ 0.0470	-0.0087 $\pm$ 0.0645	0.4845 $\pm$ 0.0394
Random Forest	75	0.5845 $\pm$ 0.1134	0.9347 $\pm$ 0.0443	-0.0404 $\pm$ 0.0240	0.4913 $\pm$ 0.0060

Table 4. Group-level ranking performance under leakage-clean DOI-held-out evaluation.

Model	N groups ranked	Hit@1	Hit@3	MRR	Unique Top-1 DES	Top-1 Diversity Ratio	Most-Common Top-1 Share
Baseline logreg full clean	119	0.983	0.991	0.9867	19	0.1597	0.2101
Hybrid true pairwise clean	119	0.958	0.991	0.9800	20	0.1681	0.2101

Table 5. Summary of group structure in the leakage-clean holdout dataset.

Metric	Value
n_groups	119
mean_group_size	1.386555
median_group_size	1
mean_n_positive	1.008403
pct_single_positive_groups	99.15966
pct_tie_for_best_groups	0.840336

Table 6. Comparison of model performance with sanity baselines under leakage-clean DOI-held-out evaluation.

Scope	Random Hit@1 (mean $\pm$ SD)	Heuristic Hit@1	Baseline Hit@1	Hybrid Hit@1	Random MRR (mean $\pm$ SD)	Heuristic MRR	Baseline MRR	Hybrid MRR
All groups	0.951 $\pm$ 0.010	0.950	0.983	0.975	0.967 $\pm$ 0.006	0.966	0.987	0.980
≥ 2 candidates	0.273 $\pm$ 0.147	0.250	0.750	0.625	0.507 $\pm$ 0.095	0.489	0.802	0.702
≥ 3 candidates	0.193 $\pm$ 0.156	0.167	0.667	0.500	0.423 $\pm$ 0.113	0.402	0.736	0.603

Random ranking represents expected performance under absence of learned structure (mean $\pm$ SD). The heuristic baseline is based on analyte-DES polarity mismatch. Limited separation between ML and heuristic performance in comparable groups indicates that ranking is largely driven by simple descriptor-level compatibility patterns.

Figures:

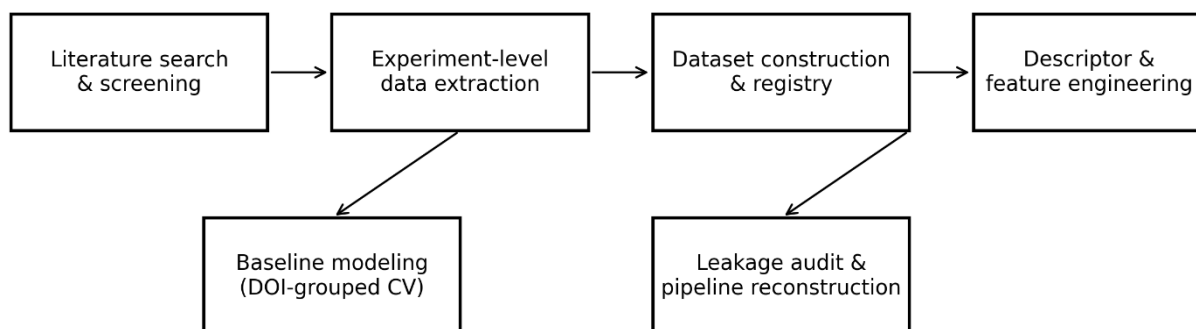


Figure 1. Overview of the data-driven workflow from literature extraction to leakage-clean model evaluation.

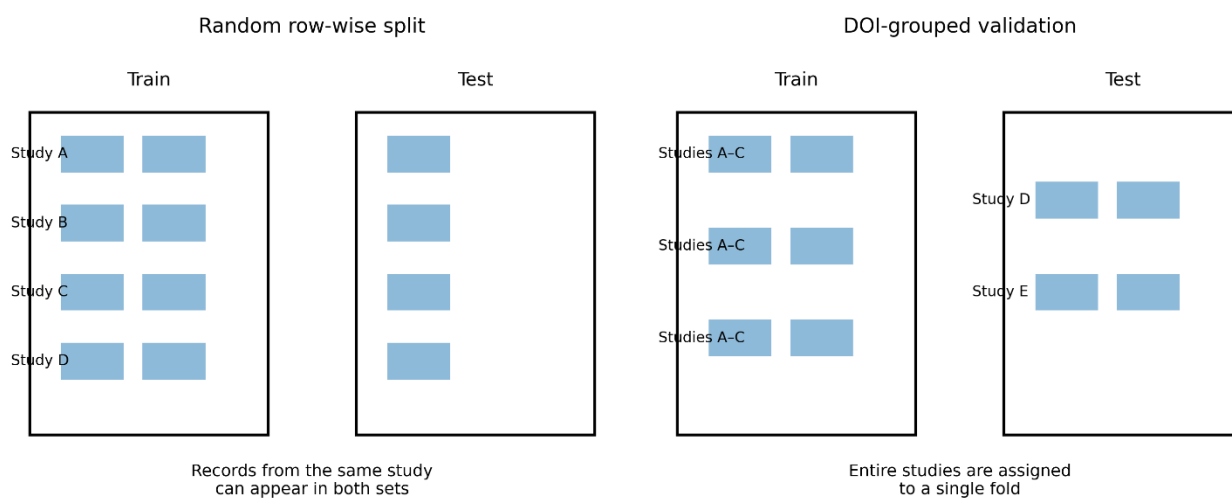


Figure 2. Comparison between random row-wise splitting and DOI-grouped validation.

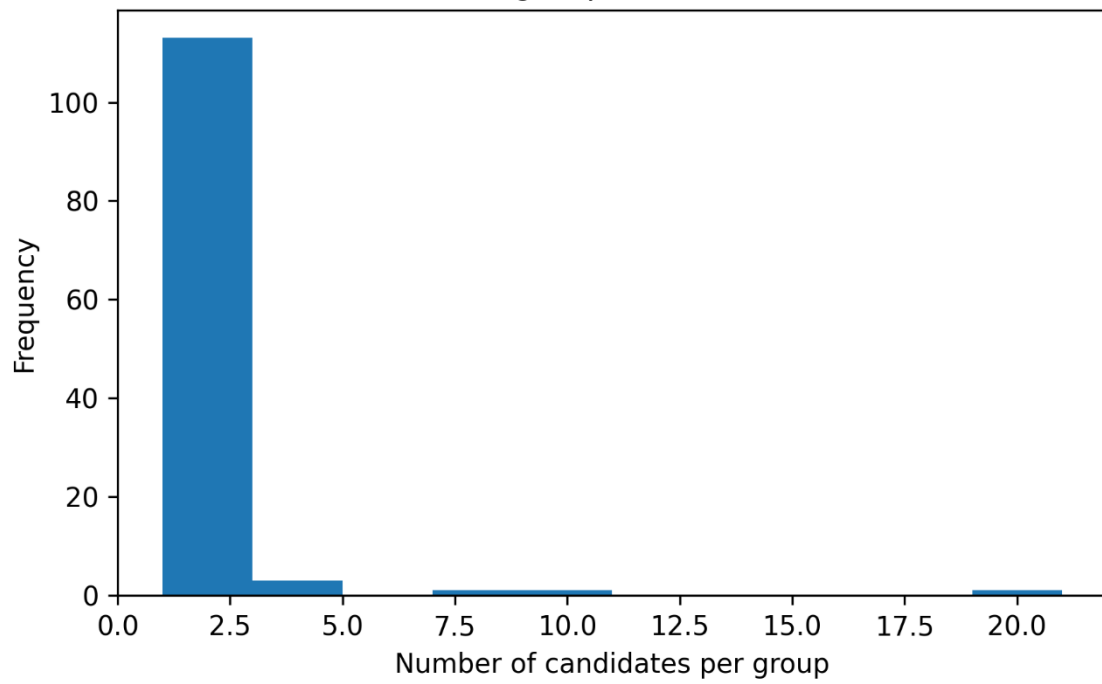


Figure 3. Distribution of group sizes (number of candidates per (DOI, analyte) group).

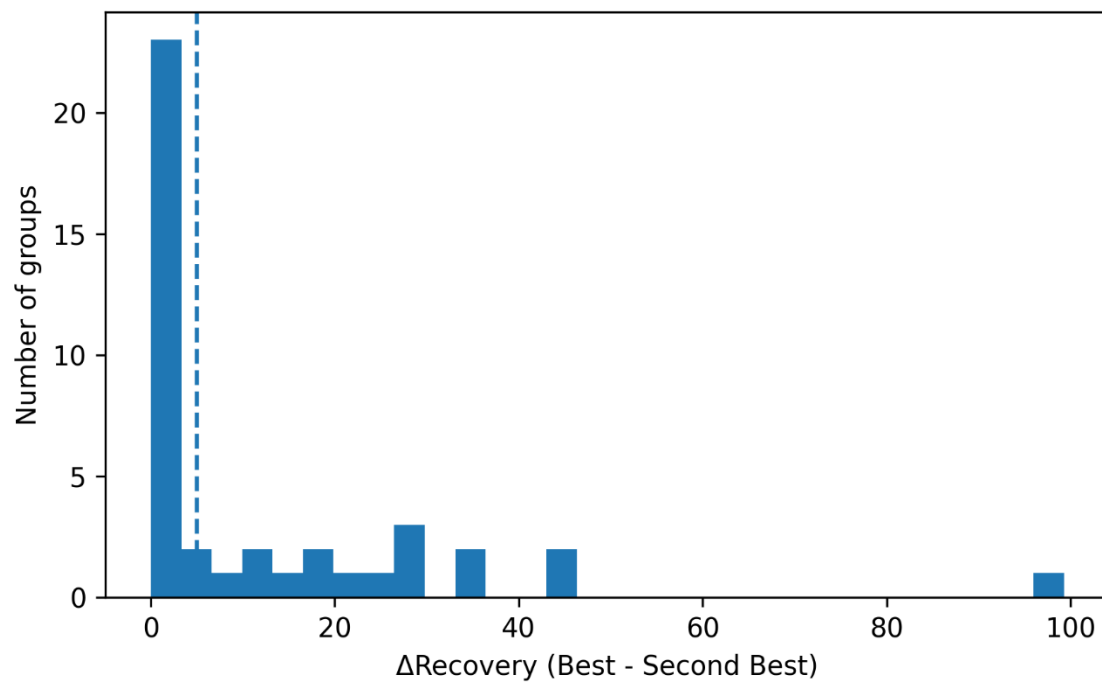


Figure 4. Distribution of recovery gaps between the best and second-best DES systems across (DOI, analyte) groups.

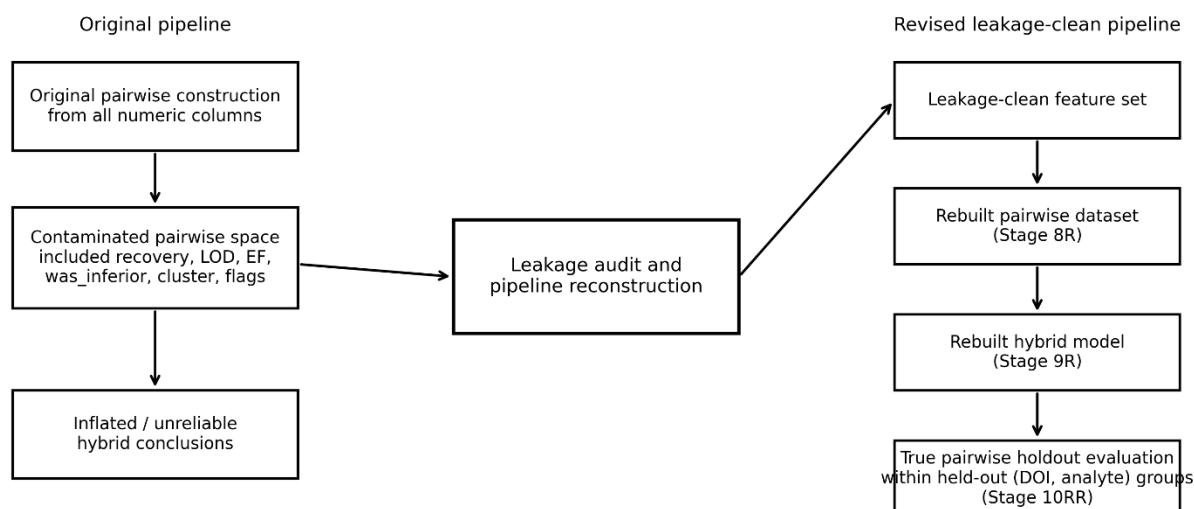


Figure 5. Reconstruction of the hybrid pipeline from contaminated pairwise features to leakage-clean true pairwise inference.

Graphical Abstract

When High Accuracy Misleads: Leakage, Degenerate Supervision, and Evaluation Fragility in Literature-Driven Machine Learning for Deep Eutectic Solvent Recommendation

A data-centric investigation showing why improved predictive accuracy does not guarantee reliable DES recommendation

